Project planning report

Saved project inputs and selected calculation results

2026-10-27 18:00:33.602389+00:00

Project

This report presents saved project inputs and the original results of the selected calculations. Country-specific requirements, electrical design and installation checks remain open. It does not approve the complete installation.

Project	Berlin · Beispielprojekt · PV und Speicher
Project reference	c7795bc2c1e14e389622015c809ace45
Input revision	28

Site and project inputs

Location	Berlin · Beispielstandort
Country	DE
Time zone	Europe/Berlin
Project note	Interaktives Beispiel mit echten JRC-Wetterdaten und nachgewiesenen Trina-Moduldaten; Dachgeometrie und Verbrauch sind Beispielannahmen.

Equipment checks

Country requirements are not assessed by the selected calculations.

Assessment status	Further checks required

Converter output

Assessment status	Checked requirements met
Result	10 kW
Source document · Page 2	10000 W https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

Inverter output limit in grid-connected operation

In grid-connected operation, the requested continuous active power from the battery cannot exceed the applicable documented maximum output of the whole inverter. Where the source gives a maximum apparent power S, $|P| \leq S$ is a necessary upper bound only; S is not an active-power rating. Passing this check does not establish continuous battery output.

Assessment status	Checked requirements met
Result	11 kW
Requirement	5 kW
Source document · Page 2	11000 VA https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

Grid phases

Assessment status	Checked requirements met
Result	3
Requirement	3

PV input power

Assessment status	Checked requirements met
Result	20 kWp
Requirement	11.04 kWp
Source document · Page 2	20000 Wp https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

PV strings and inputs

Assessment status	Further checks required

Required accessory

Assessment status	Checked requirements met
Result	SAJ BC2-HV1

Manufacturer pairing

Assessment status	Further checks required

Module count

Assessment status	Checked requirements met
Result	4
Requirement	2 – 5

Parallel arrangement

Assessment status	Checked requirements met
Result	1

Battery voltage

Assessment status	Checked requirements met
Source document · Page 2	180~800 V https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf
Source document · Page 2	89.6-115.2 / 179.2-230.4 / 268.8-345.6 / 358.4-460.8 / 448-576.0 V https://de-linker.com/wp-content/uploads/2026/09/B2-5-25K-HV1HV5-datasheet-EN-BEU.pdf

Sales unit

Assessment status	Checked requirements met
Source document · Page 2	Battery module https://de-linker.com/wp-content/uploads/2026/09/B2-5-25K-HV1HV5-datasheet-EN-BEU.pdf

Usable energy

4.5 × 4 = 18 kWh

Assessment status	Checked requirements met
Result	18 kWh
Requirement	10 kWh

Source document · Page 2	4.5 kWh https://de-linker.com/wp-content/uploads/2026/09/B2-5-25K-HV1HV5-datasheet-EN-BEU.pdf

Continuous AC discharge power

Assessment status	Further checks required
Requirement	5 kW

Battery discharge path

Assessment status	Further checks required

Continuous AC charge power

Assessment status	Further checks required
Requirement	5

Components and manufacturer conditions

The H2 T3 ships with a smart meter, three current transformers and a 10 m meter communication cable (H2 manual, section 2.2 Unpacking, PDF p. 36 English / p. 7 German).

SAJ's B2 datasheet names B2-5.0 to B2-25.0 as one to five BU2-5.0 modules with one BC2-HV1 control module, and the B2 manual (V1.2, PDF p5) says the B2 can only be used as such a set: the control module is an article of its own.

SAJ's Configuration Instructions V1.0 (Section 2) give the H2 series system as H2 + BC2 + BU2. SAJ's B2 datasheet names B2-5.0 to B2-25.0 as one to five BU2-5.0 modules with one BC2-HV1 control module, and SAJ's B2 manual (V1.2, PDF p5) says the B2 can only be used as a set with SAJ's H2 high voltage series storage inverter. The listing does not cover the one-module B2-5.0, whose operating voltage lies below the H2 T3's battery voltage range.

One BU2-5.0 module alone operates at 89.6 to 115.2 V (SAJ's B2 datasheet gives this for the one-module B2-5.0), below the H2 T3's battery voltage range of 180 to 800 V; two modules operate at 179.2 to 230.4 V.

Source document · Page 36	https://de-linker.com/wp-content/uploads/2026/10/SAJ_H2_10-30K_User-Manual_DE-EN_EU.pdf https://de-linker.com/wp-content/uploads/2026/10/SAJ_H2_10-30K_User-Manual_DE-EN_EU.pdf
Source document	https://de-linker.com/wp-content/uploads/2026/09/H2_HS2_HS3_AS2_AS3_Configuration_Instructions_V1.0_en.pdf https://de-linker.com/wp-content/uploads/2026/09/H2_HS2_HS3_AS2_AS3_Configuration_Instructions_V1.0_en.pdf
Source document	https://de-linker.com/wp-content/uploads/2026/10/B2-5.0-HV1-HV5-user-manual-EN-EU.pdf https://de-linker.com/wp-content/uploads/2026/10/B2-5.0-HV1-HV5-user-manual-EN-EU.pdf
Source document · Page 2	https://de-linker.com/wp-content/uploads/2026/09/B2-5-25K-HV1HV5-datasheet-EN-BEU.pdf https://de-linker.com/wp-content/uploads/2026/09/B2-5-25K-HV1HV5-datasheet-EN-BEU.pdf
Source document · Page 37	https://de-linker.com/wp-content/uploads/2026/10/SAJ_H2_10-30K_User-Manual_DE-EN_EU.pdf https://de-linker.com/wp-content/uploads/2026/10/SAJ_H2_10-30K_User-Manual_DE-EN_EU.pdf

Provisional bill of materials

This list combines the selected equipment, saved roof modules and documented included components. Quantities are shown where established. Mounting, cabling, protection and other installation materials still require project-specific design.

Source document · Page 36

Manufacturer package contents
https://de-linker.com/wp-content/uploads/2026/10/SAJ_H2_10-30K_User-Manual_DE-EN_EU.pdf

Quantity	Product	Reference	Supply scope
1 pc	SAJ H2-10K-T3	DLK-SAJ-H210KT3	Selected equipment
5 pc	M5×12 screw assembly		Included with the equipment · SAJ H2-10K-T3
6 pair	PV connector pair (T3)		Included with the equipment · SAJ H2-10K-T3
2 pc	2-pin pluggable terminal block		Included with the equipment · SAJ H2-10K-T3
1 pc	3-pin pluggable terminal block		Included with the equipment · SAJ H2-10K-T3
1 pc	6-pin pluggable terminal block		Included with the equipment · SAJ H2-10K-T3
1 pc	Communication terminal cover		Included with the equipment · SAJ H2-10K-T3
1 pc	Inverter mounting bracket		Included with the equipment · SAJ H2-10K-T3
2 pc	Positive battery connector		Included with the equipment · SAJ H2-10K-T3
2 pc	Negative battery connector		Included with the equipment · SAJ H2-10K-T3
10 pc	AC terminal		Included with the equipment · SAJ H2-10K-T3
4 pc	M6×50 screw assembly		Included with the equipment · SAJ H2-10K-T3
1 pc	eSolar AIO3		Included with the equipment · SAJ H2-10K-T3
1 pc	AC terminal cover		Included with the equipment · SAJ H2-10K-T3
1 pc	Smart meter		Included with the equipment · SAJ H2-10K-T3
3 pc	Current transformer		Included with the equipment · SAJ H2-10K-T3
1 pc	Meter communication cable (10 m)		Included with the equipment · SAJ H2-10K-T3
10 pc	RJ45 plug		Included with the equipment · SAJ H2-10K-T3
1 pc	SAJ BC2-HV1	DLK-SAJ-BC2HV1	Selected equipment
4 pc	SAJ BU2-5.0-HV1	DLK-SAJ-BU250HV1	Selected equipment
24 pc	Trina Solar TSM-460NEG9R.28		PV reference module · project quotation

PV modules used in the calculation

Module counts come from the saved layout of the selected generation result. The listed modules are design references; electrical and mounting suitability require project-specific assessment.

Input revision	26
PV nameplate power	11.04 kWp
Source document · Page 2	Trina Solar TSM-460NEG9R.28 https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S

_NEG9R.28_EN_2024_C_web.pdf

Quantity	Model	Module power (STC)
24	Trina Solar TSM-460NEG9R.28	460 W

Energy scenario and selected equipment

These checks compare the saved scenario parameters with the equipment selected for this calculation. They do not establish manufacturer performance.

4.5 × 4 = 18 kWh

Assessment status	Further checks required
Model	SAJ BU2-5.0-HV1
Model	SAJ BC2-HV1
Model	SAJ H2-10K-T3
Module count	4
Parallel arrangement	1
Usable energy · Source document · Page 2	4.5 kWh https://de-linker.com/wp-content/uploads/2026/09/B2-5-25K-HV1HV5-datasheet-EN-BEU.pdf

Check	Assessment status	Result	Requirement
Usable energy	Checked requirements met	18 kWh	10 kWh
Continuous AC discharge power	Further checks required	—	5 kW
Continuous AC charge power	Further checks required	—	5 kW
Scenario operating mode	Checked requirements met	—	—
Storage losses and operating conditions	Further checks required	—	—
Converter ports and system topology	Further checks required	—	—

PVGIS roof AC reference

PVGIS models AC generation for the saved roof in the stated calendar year, using the stated reference technology, mounting type and system loss setting. Device-specific inverter behavior, nearby obstacles and shading between module rows are outside this model. Electrical design and grid acceptance require separate assessment.

Calendar year	2023
Radiation database	PVGIS-SARAH3
PV reference technology	crystSi
PVGIS mounting type	Free-standing (free)
PVGIS system loss setting	14 %
Terrain horizon	Omitted
Reference model assumption note	Explicit synthetic JRC calendar-year AC reference: SARAH3 2023, crystSi, free-standing mounting, 14% generic system-loss setting and no terrain horizon. The 14% is the named generic JRC scenario assumption, not a measured roof loss or manufacturer/device performance. Saved Berlin city-anchor roof and 6000 kWh/year load are synthetic; downstream storage calculation remains 16 June 2023, 09:00–11:00 UTC.
Modelled AC generation	11,008.95 kWh
Calculation period (UTC)	2023-01-01T00:00:00+00:00 – 2024-01-01T00:00:00+00:00
Calculated intervals	8,760

Source calculation / input revision	0029fbb5b0ac475196233a6726cb2dad / 28
Source calculation / input revision	8963c93a7781405b8bc0c83e672da43a / 26
PVGIS source for array	array https://re.jrc.ec.europa.eu/api/v5_3/seriescalc?lat=52.52&lon=13.405&raddatabase=PVGIS-SARAH3&startyear=2023&endyear=2023&pvcalculation=1&peakpower=11.04&loss=14.0&pvtechchoice=crystSi&mountingplace=free&angle=30.0&aspect=0.0&trackingtype=0&usehorizon=0&components=1&optimalinclination=0&optimalangles=0&outputformat=json

Energy scenario

These values come from the selected saved energy result and reflect its original scenario parameters. The general energy-flow model does not establish the performance of the selected equipment.

Calculation period (UTC)	2023-06-16T09:00:00+00:00 – 2023-06-16T11:00:00+00:00
Simulated load demand	1.103 kWh
PV generation	4.316 kWh
Grid import	0 kWh
PV export	0 kWh
PV curtailment	0 kWh
Unserved simulated load	0 kWh
Solar share of gross demand	100 %
PV used or stored on site	100 %
Calculated intervals	2

Saved calculation references

Data source	Beispielannahme · kein externer AC-Hilfsverbrauch
Data source	Deklariertes Beispiel: 10 kWh nutzbar, 5 kW AC, je Richtung 96 % Wirkungsgrad, keine Hilfs- oder Standverluste. Kein Produktdatenblatt und keine Gerätefreigabe.
Data source	Beispielverbrauch · 6.000 kWh pro Jahr
Data source	JRC PVGIS calendar-year roof AC reference
Calculation reference	0029fbb5b0ac475196233a6726cb2dad · Input revision 28
Calculation reference	8963c93a7781405b8bc0c83e672da43a · Input revision 26

Certificates and related product documents

These documents were linked to the selected products when this result was recorded. Their applicability to this project’s models, firmware, configuration, standard editions and annexes has not been assessed here. Compliance with country-specific requirements, installation approval and grid-operator acceptance require separate evidence.

BU2-5.0-HV1	SAJ B2 battery systems and BU2-5.0 battery modules: EU declaration of conformity (multilingual) https://de-linker.com/wp-content/uploads/2026/10/SAJ_B2_BU2-5.0_EU-Declaration-of-Conformity.pdf?linker_document_sha256=6a45702bfe4ae9de06c173b084b3decbf6ac9cc3cc5f77cbe0f8686885c9ad6e
SHA-256	6a45702bfe4ae9de06c173b084b3decbf6ac9cc3cc5f77cbe0f8686885c9ad6e
Recorded document revision	cead73235a8d8b291c22189e9f589c4d6f5c89dd45383599ac97c500bbceb4fb
BC2-HV1	SAJ B2 battery systems and BC2-HV1 control module: EU declaration of conformity (multilingual) https://de-linker.com/wp-content/uploads/2026/10/SAJ_B2_BC2-HV1_EU-Declaration-of-Conformity.pdf?linker_document_sha256=9cb10f8699f660cbbde4197f3

	0b392b8b7b77d95723a8a0ad4c18a416b854a61
SHA-256	9cb10f8699f660cbbde4197f30b392b8b7b77d95723a8a0ad4c18a416b854a61
Recorded document revision	dd94420889a64b164722a24f947f6e4404df7e99739b6741b35b686639448773
H2-10K-T3	EU declaration of conformity for the SAJ H2 inverters, 13 languages including English and German (names the H2-25K-T3) https://de-linker.com/wp-content/uploads/2026/10/SAJ_H2_EU-Declaration-of-Conformity.pdf?linker_document_sha256=9881d06dabba6e53893923e93ebb90365b966429967104cb92890de39573b443
SHA-256	9881d06dabba6e53893923e93ebb90365b966429967104cb92890de39573b443
Recorded document revision	77800e74a3c6fff9de321ad9a6225955df5e7d863eaa392bfdd52598a245cf45